

Scalable Airport Network Infrastructure Design

A hierarchical, VLAN-segmented network architecture for a multi-terminal airport campus — covering IP addressing, routing, inter-VLAN switching, core network services, and wireless connectivity for staff and security operations.

Project: AeroNova — A Scalable Airport Network

Platform: Cisco Packet Tracer

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AeroNova Airport Network — Technical Design Document

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01 Executive Summary

AeroNova is a hierarchical enterprise network designed to support the full operational footprint of a modern airport terminal — administrative management on both wings of the building, a centralized server room, and three public-facing passenger halls — while keeping each functional area logically isolated, independently addressable, and centrally routable.

The design follows a classic **core–distribution–access** model built entirely in Cisco Packet Tracer. Three ISR4331 routers form the routed core, interconnected over point-to-point serial links, while a Catalyst 3650 multilayer switch performs Layer 3 inter-VLAN routing across the airport's three passenger halls. A dedicated server room hosts the six centralized network services — DHCP, DNS, Web, FTP, Mail, and NTP — that every device on the network relies on, and each hall additionally runs its own local web server for hall-specific flight information.

The addressing plan uses the private range **10.20.0.0/16**, subnetted into six /24 blocks — one per functional zone — which keeps the scheme simple to document, easy to troubleshoot, and comfortably scalable, since each subnet supports up to 254 hosts. Wireless access points extend connectivity to management staff (laptops, tablets, smartphones) and to the security team's mobile devices throughout the terminal, without requiring a wired drop for every device.

3 × ISR4331 Routers

1 × Catalyst 3650 Multilayer Switch

6 × 2960 Access Switches

6 Core Network Services

6 Subnets / VLANs

Wireless Access for Staff & Security

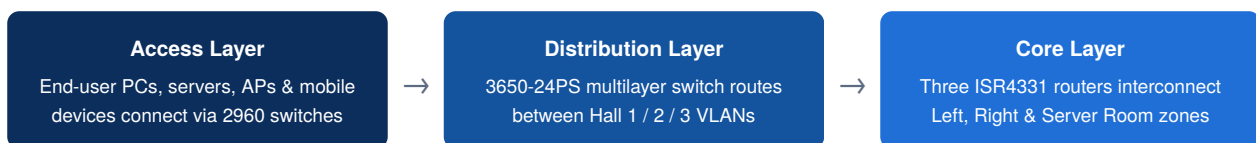
02 Project Objectives

The AeroNova network was designed to satisfy the operational and technical requirements of a mid-sized airport terminal. The core objectives guiding every design decision were:

- **Logical segmentation** — separate administrative, server, and passenger-facing traffic into distinct broadcast domains so that a problem or a security incident in one zone cannot directly affect another.
- **Scalable, predictable addressing** — use a consistent, easy-to-read IP scheme that leaves headroom for future halls, gates, or services to be added without re-numbering the existing network.
- **Centralized, always-available services** — provide DHCP, DNS, web hosting, file transfer, email, and time synchronization from a single, well-secured server room reachable from every part of the network.
- **Resilient inter-site routing** — connect the left wing, the right wing, and the server core through a routed backbone so that every subnet can reach every other subnet.
- **Passenger hall autonomy with shared reach** — give each hall its own VLAN and local web server for hall-specific content, while still allowing controlled inter-VLAN communication through the multilayer switch.
- **Mobility for staff and security** — extend wireless coverage so administrative staff and the security team can stay connected while moving between halls and management areas.

03 Network Architecture Overview

The network is organized around three logical tiers. A **routed core** made up of three ISR4331 routers handles inter-site traffic between the left management wing, the right management wing, and the central server room. Hanging off the core is a **distribution layer** — a Catalyst 3650 multilayer switch — that performs Layer 3 inter-VLAN routing for the three passenger halls. Finally, an **access layer** of 2960 series switches and wireless access points connects the actual end devices: PCs, servers, laptops, tablets, and smartphones.



Design Rationale

Why a routed core?

Point-to-point serial links between the three ISR4331 routers keep the management wings and the server room as independently routable zones, so traffic between them is explicitly controlled at Layer 3 rather than flooded across a flat Layer 2 network.

Why a multilayer switch for the halls?

Passenger halls generate the highest device density. A Layer 3 switch lets each hall keep its own VLAN and subnet for isolation, while still routing between halls at switching speed instead of hair-pinning traffic through a router.

Why a centralized server room?

Placing DHCP, DNS, Web, FTP, Mail, and NTP in one physically and logically separate subnet (10.20.30.0/24) simplifies security policy, backup, and monitoring, and gives every other zone a single, predictable path to core services.

Why wireless at the edges?

Access points in the management wings and passenger halls let staff laptops/tablets and the security team's smartphones roam the terminal freely, without depending on fixed cable runs to every possible location.

04 Network Topology Diagram

The diagram below is the full logical topology as implemented in Cisco Packet Tracer. It shows the three-router core, the central server room, the left and right management zones, the multilayer-switched hall access layer, and the wireless access points serving mobile devices throughout the terminal.

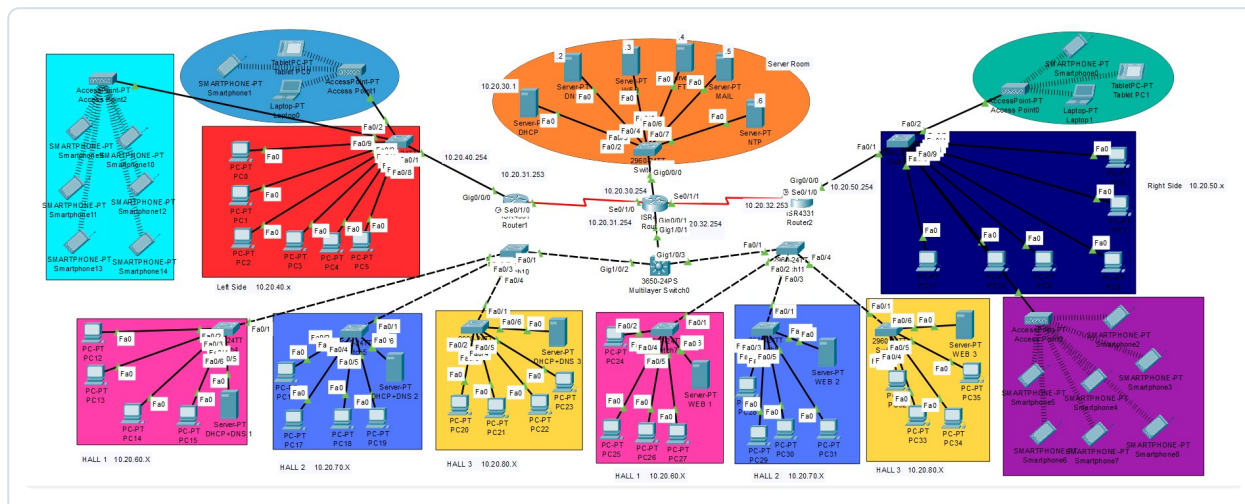


Figure 1 — AeroNova full network topology (Cisco Packet Tracer)

Reading the Topology

- **Center (orange):** Server Room — DNS, Web, DHCP, FTP, Mail and NTP servers, connected through a 2960 switch to the core router's Gig0/0/0 interface.
- **Left (red):** Left Side Management zone (10.20.40.x) — administrative PCs, connected to Router1 and served by an access point for laptops, a tablet, and staff smartphones.
- **Right (dark blue):** Right Side Management zone (10.20.50.x) — mirrors the left wing, connected to Router2 with its own access point for mobile management devices.
- **Bottom (pink / blue / yellow / purple):** Halls 1, 2 and 3, each with its own VLAN, access switch, dedicated web server, and — in several locations — wireless access for the security team's smartphones.
- **Center-bottom:** the 3650-24PS multilayer switch, which ties the hall access switches together and routes between hall VLANs and up to the core router.

05 Physical Topology

Alongside the logical topology, the project also models the airport's **physical layout** — the actual buildings and the cable runs between them overlaid on the site map. This view confirms that the logical design (VLANs, subnets, routing) maps onto a realistic, buildable campus layout rather than an abstract diagram only.

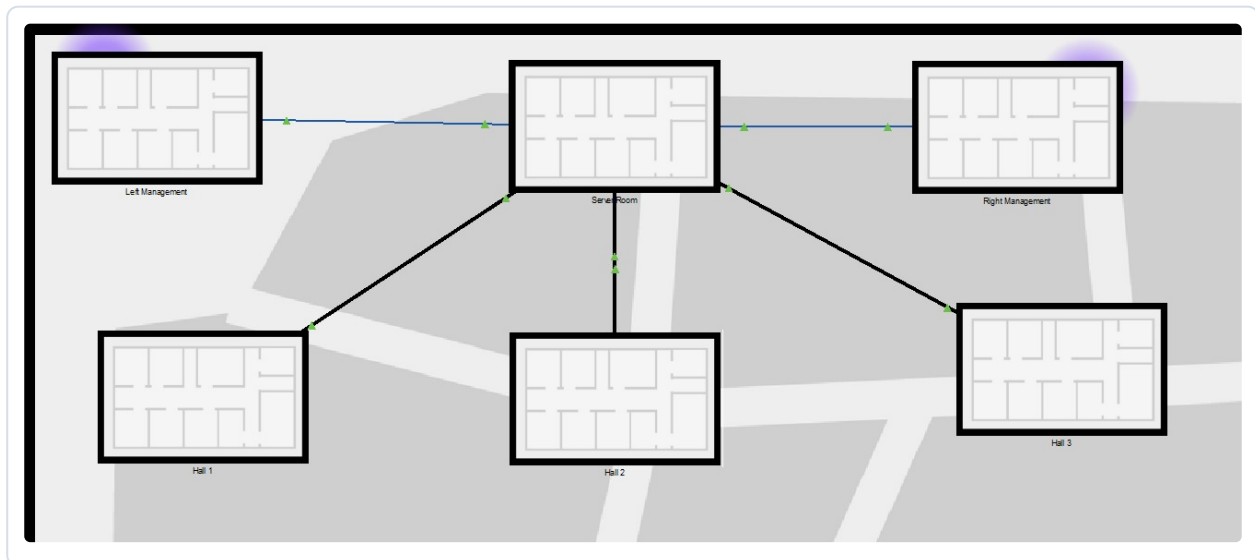


Figure 2 — AeroNova physical topology overlaid on the site layout (Cisco Packet Tracer)

Reading the Physical Layout

- **Server Room** sits centrally on the site and is the physical hub of the campus, with direct cable runs out to the Left Management and Right Management buildings and down to Hall 2.
- **Left Management** and **Right Management** buildings flank the Server Room on either side, each connected back to it by a dedicated link — matching the logical design where each management zone routes through the core.
- **Hall 1, Hall 2 and Hall 3** are physically separate buildings south of the Server Room; Hall 1 and Hall 3 connect back to the Server Room directly, while Hall 2 sits on the direct vertical run beneath it — mirroring the star-shaped cabling pattern used in the logical topology.

06 Device Inventory

The table below summarizes the active infrastructure devices modeled in the project.

Category	Device	Qty	Role
Routing	Cisco ISR4331	3	Core routing between Server Room, Left Management and Right Management zones over serial WAN links
Switching (L3)	Catalyst 3650-24PS	1	Distribution-layer multilayer switch; inter-VLAN routing for Hall 1 / 2 / 3
Switching (L2)	Catalyst 2960-24TT	6	Access switches for the Server Room, Left zone, Right zone, and hall access points
Wireless	Access Point (802.11)	4+	Wireless coverage for management laptops/tablets and hall/security smartphones
Servers	Server-PT (DHCP, DNS, Web, FTP, Mail, NTP)	6	Centralized network services in the Server Room
Servers	Server-PT (Hall Web Servers)	3	Local, hall-specific web/flight-information hosting (one per hall)
Endpoints	PC-PT (desktops)	36	Staff and hall workstations across all zones
Endpoints	Laptop-PT / Tablet-PT	4	Wireless management-staff devices
Endpoints	Smartphone-PT	15	Wireless devices for hall coverage and the security team

Quantities reflect the device count visible in the Packet Tracer topology and may vary slightly by final build revision.

07 IP Addressing & Subnetting Scheme

Every functional zone in the airport is assigned its own /24 subnet from the private **10.20.0.0/16** range, using the third octet as a location identifier. This keeps the scheme logical, scalable, and secure: 254 usable host addresses per zone is comfortably more than any single room or hall currently needs, while leaving the fourth octet free for structured host assignment (servers at the low end, DHCP pools in the middle, and the gateway fixed at .254).

Zone / Subnet	Network Address	Mask	Default Gateway	Purpose
Server Room	10.20.30.0	/24	10.20.30.254	DHCP, DNS, Web, FTP, Mail, NTP servers
Left Management	10.20.40.0	/24	10.20.40.254	Left-wing administrative PCs & wireless devices
Right Management	10.20.50.0	/24	10.20.50.254	Right-wing administrative PCs & wireless devices
Hall 1	10.20.60.0	/24	10.20.60.254	Passenger Hall 1 workstations, web server & wireless
Hall 2	10.20.70.0	/24	10.20.70.254	Passenger Hall 2 workstations, web server & wireless
Hall 3	10.20.80.0	/24	10.20.80.254	Passenger Hall 3 workstations, web server & wireless

Server Room Host Assignment

Within the 10.20.30.0/24 subnet, servers are assigned fixed low-range addresses so they are easy to locate and reference from DNS or documentation:

Address	Server	Function
10.20.30.1	DHCP Server	Automatic IP assignment for client subnets
10.20.30.2	DNS Server	Name resolution for the aeronova.local domain
10.20.30.3	Web Server	Hosts the main AeroNova site
10.20.30.4	FTP Server	Secure file transfer between users
10.20.30.5	Mail Server	Internal email service
10.20.30.6	NTP Server	Time synchronization for logging & security events
10.20.30.254	Core Router (Gig0/0/0)	Default gateway for the Server Room

Point-to-point WAN links between the core router and Router1 / Router2 use small dedicated transit subnets (10.20.31.0/30 and 10.20.32.0/30) that are excluded from the host-facing addressing table above, since they carry no end-user traffic.

08 VLAN Segmentation Plan

Each passenger hall is mapped one-to-one to its own VLAN and IP subnet, isolating hall traffic from the management and server zones while still allowing controlled routing between them through the network's Layer 3 devices.

VLAN	Name	Subnet	Access Switch(es)	Notes
—	Server Room	10.20.30.0/24	2960-24TT (Server Room)	Routed directly off the core router; not a switched VLAN in the hall sense
—	Left Management	10.20.40.0/24	2960-24TT + AP (Left)	Wired PCs plus wireless AP for laptop/tablet/smartphone access
—	Right Management	10.20.50.0/24	2960-24TT + AP (Right)	Mirrors the left management zone
VLAN 60	Hall 1	10.20.60.0/24	2960-24TT (Hall 1 access)	Local Web Server 1 + inter-VLAN routing via 3650
VLAN 70	Hall 2	10.20.70.0/24	2960-24TT (Hall 2 access)	Local Web Server 2 + inter-VLAN routing via 3650
VLAN 80	Hall 3	10.20.80.0/24	2960-24TT (Hall 3 access)	Local Web Server 3 + inter-VLAN routing via 3650

VLAN numbers follow the third octet of each hall's subnet (e.g. Hall 1 → 10.20.60.0/24 → VLAN 60) for a naming convention that is self-documenting during troubleshooting.

09 Core Routing & WAN Interconnection

The three ISR4331 routers form a routed hub: the central router sits between Router1 (left) and Router2 (right), joined by dedicated point-to-point serial links. This keeps each zone's broadcast traffic contained locally while giving every subnet a deterministic path to every other subnet through the core.

Router	Interface	IP Address	Connects To
Core Router	Gig0/0/0	10.20.30.254	Server Room switch
Core Router	Se0/1/0	10.20.31.254	Router1 (Left) — serial WAN link
Core Router	Se0/1/1	10.20.32.254	Router2 (Right) — serial WAN link
Core Router	Gig1/0/1	—	3650-24PS Multilayer Switch (Gig1/0/3) — hall VLAN routing
Router1 (Left)	Gig0/0/0	10.20.40.254	Left Management switch
Router1 (Left)	Se0/1/0	10.20.31.253	Core Router — serial WAN link
Router2 (Right)	Gig0/0/0	10.20.50.254	Right Management switch
Router2 (Right)	Se0/1/0	10.20.32.253	Core Router — serial WAN link

Static or dynamic routes (depending on final configuration) on each router advertise the local management or server subnet across the serial links, so that, for example, a PC in the Left Management zone can reach the Mail Server in the Server Room via Router1 → Core Router, and a PC in the Right Management zone can reach the same server via Router2 → Core Router.

10 Inter-VLAN Routing & Hall Web Services

Each passenger hall is assigned to its own VLAN and runs its own dedicated web server that hosts hall-specific flight information. The 3650-24PS multilayer switch performs **inter-VLAN routing** between the three halls, so a device in Hall 1 can, when authorized, reach the Hall 2 or Hall 3 web server — while normal traffic stays logically separated by default. This gives the design both **segmentation** (each hall is its own broadcast domain and subnet) and **seamless reachability** (any hall can still reach shared or cross-hall resources through the routed distribution layer).

Hall Web Server 1 — 10.20.60.x

Hosts flight and facility information local to Hall 1; reachable from other halls via the multilayer switch's inter-VLAN routes.

Hall Web Server 2 — 10.20.70.x

Hosts flight and facility information local to Hall 2, following the same access model as Hall 1.

Hall Web Server 3 — 10.20.80.x

Hosts flight and facility information local to Hall 3, completing the per-hall web service pattern used across the terminal.

This pattern keeps hall-specific content close to the users who need it most (reducing traffic that would otherwise cross the core), while the multilayer switch's routed SVIs ensure no hall is an isolated island — a passenger or staff device in any hall can still resolve and reach services in another.

11 Network Services

The Server Room centralizes six core services that every zone in the network depends on.

Service	Address	Description
DHCP	10.20.30.1	Automatically assigns IP addresses and configuration parameters — subnet mask, gateway, DNS server — to clients in each zone.
DNS	10.20.30.2	Translates domain names into IP addresses for every host on the network.
Web	10.20.30.3	Hosts the main AeroNova site, accessible through standard web browsers.
FTP	10.20.30.4	Provides secure file transfer and sharing between authorized users.
Mail	10.20.30.5	Enables users across the network to send and receive internal email.
NTP	10.20.30.6	Synchronizes device date and time for accurate logging, monitoring, and security-event timestamps.

Because these services live in one dedicated subnet reachable through the routed core, every zone — Left Management, Right Management, and all three halls — resolves names, obtains addresses, and reaches shared resources through the same predictable path, simplifying both day-to-day operations and troubleshooting.

12 Wireless Connectivity

Wireless access points extend the wired network to mobile staff and security devices in three main areas: the Left and Right Management zones, and the passenger halls.

Management Subnets

Wireless access in the Left and Right Management zones gives administrative staff flexible, reliable connectivity for laptops, tablets, and smartphones — letting them reach internal management resources without depending entirely on a wired port, and supporting mobility between different areas of the terminal while staying on their assigned subnet.

Smart Areas & Security Mobility

Additional wireless coverage across the hall areas — referred to in the design as Smart Areas — supports the larger number of mobile devices carried by the airport's security team. This lets security personnel remain connected and communicate as they move between halls and different parts of the terminal, coordinating quickly across areas rather than being tied to a single fixed access point.

Security Team Communication

The wireless network is specifically relied on to let security personnel communicate and coordinate in real time across every hall and management area of the airport, which is a core operational requirement for a live terminal environment.

13 Security Considerations

The segmentation already built into AeroNova's design — separate subnets per zone, VLAN isolation per hall, and a dedicated server room — forms the foundation of a defensible network. Device management access control was implemented directly in the project, and the points below cover both what is configured today and what would extend the design toward a production-grade deployment.

Implemented in This Project

- **SSH for remote device management** — Telnet is disabled and SSH is configured on the routers and switches, so administrators authenticate with encrypted credentials instead of plain-text remote access.

- **Access control lists (ACLs) on specific hosts** — targeted ACLs restrict which hosts are permitted to reach sensitive devices/services, rather than leaving those paths open to the entire subnet.

Recommended for a Production Deployment

- **Port security** on access-layer switches to limit the number of MAC addresses per port and prevent rogue devices from joining a hall or management VLAN.
- **Wireless authentication** (WPA2/WPA3-Enterprise) on every access point serving management and security devices, rather than a shared pre-shared key.
- **Restricting SSH access further** by source subnet (management zones only) and enforcing strong, unique per-device credentials on top of the SSH access already in place.
- **Centralized logging** — pair the NTP server's accurate timestamps with a syslog target so security and network events across all zones can be correlated reliably.
- **Redundant WAN paths** — consider a secondary link between Router1 and Router2 (bypassing the core router) so a single serial-link failure does not isolate a management zone from the rest of the network.

14 Conclusion

AeroNova demonstrates a complete, working model of an airport-scale network built on well-established enterprise design principles: a routed core, a Layer 3 distribution switch for high-density areas, VLAN-segmented access, and centralized network services. The 10.20.0.0/16 addressing scheme — one /24 per functional zone, gateway fixed at the top of the range — keeps the network easy to document, extend, and troubleshoot as the airport grows.

Together, the routed core, the hall-level VLANs with inter-VLAN routing, the centralized services in the Server Room, and the wireless coverage for staff and security mobility give AeroNova the segmentation, scalability, and operational reach expected of a real airport network — while remaining simple enough to manage, monitor, and extend.

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